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# Style

# Truth and Nil: Niladic

## Wuxia

俠

*The Hero*

U+4FA0

## Nil



*The empty set — quiet failure value; absorbs; Nil⊡function → Nil*

U+2205

## Truth

⊤

*Verum — first-class truth; no true anymore*

U+22A4

# Truth and Nil: Monadic

## Invert

~

*Invert — monadic only (tilde). Scalar: ~ 0 → 1, ~ 1 → 0 (user wording to confirm). Unary scalar verbs operate on array back ranks (new 2026-10-11)*

U+007E

## Not

¬

*Not — monadic boolean not: ¬ Truth yields Nil, ¬ Nil yields Truth (new 2026-10-14)*

U+00AC

## Congruent

≡

*Ambivalent: monadic binds train to lhs⊡function*

U+2261

# Truth and Nil: Dyadic

## Less Than



*Less Than — dyadic < A B yields Truth when A < B, Nil otherwise (new 2026-10-14)*

U+003C

## Greater Than



*Greater Than — dyadic > A B yields Truth when A > B, Nil otherwise (new 2026-10-14)*

U+003E

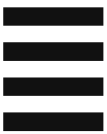
## Congruent



*Ambivalent: dyadic ≡ A B yields Truth when A and B have the same binding (same location in the system tables), Nil otherwise*

U+2261

## IsAt



*IsAt — dyadic ≣ X Y yields Truth when X and Y share the same pointer address, Nil otherwise (new 2026-10-14)*

U+2263

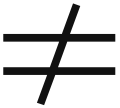
## Equality



*Equality — dyadic = A B yields Truth when A = B, Nil otherwise (new 2026-10-14)*

U+003D

## Unequal



*Unequal — dyadic ≠ A B: sugar for ¬ = A B (boolean Not of Equality) (new 2026-10-14)*

U+2260

## Less Or Equal



*Less Or Equal — dyadic ≤ A B yields Truth when A ≤ B, Nil otherwise (new 2026-10-14)*

U+2264

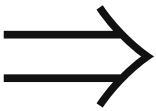
## Greater Or Equal



*Greater Or Equal — dyadic ≥ A B yields Truth when A ≥ B, Nil otherwise (new 2026-10-14)*

U+2265

## Implies



*Implies — logical implication; ∅ ⇒ ∅ → ⊤*

U+21D2

## forall



*Train quantifier — every element must evaluate non-nil. Idiom: dyadic NAND is ¬ ∀ 𝕀𝕀 X Y*

U+2200

## Approx



*Approx — dyadic ≈ A B yields Truth when|A - B|is within 1e-6, Nil otherwise (new 2026-10-14)*

U+2248

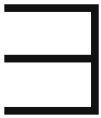
## UnApprox



*UnApprox — dyadic ≉ A B: sugar for ¬ ≈ A B (boolean Not of Approx) (new 2026-10-14)*

U+2249

## Forsome



*Existential — at least one element evaluates non-nil*

U+2203

## Fornone



*Not-forsome — defined as ~Forsome*

U+2204

# Arithmetic: Niladic

## Signum / Plus

+

*Ambivalent: Niladic + yields 0.*
*(Was ∓ U+2213, retired 2026-10-12*

U+002B

## Negate / Subtract

−

*Niladic - yields 0 (new 2026-10-11*

U+002D

## Plus (Complex)

⨢

*Plus (Complex), aka Sum Complex, Add Complex —; Niladic: sugar for ℿ 0 0, the complex zero (new 2026-10-14)*

U+2A22

## Times (Complex)

⨣

*Times (Complex) —; Niladic: sugar for ℿ 1 0, the complex one (new 2026-10-14)*

U+2A30

## Minus (Complex)



*Minus (Complex) —; Niladic: sugar for ⨿ 0 0, the complex zero (new 2026-10-14)*

U+235B

## Times (Polar)



*Times (Polar) —; Niladic: the polar one, 2×1 [1; 0] (new 2026-10-14)*

U+2A02

## Divide (Complex)



*Divide (Complex) —; Niladic: sugar for ⨿ 1 0, the complex one (new 2026-10-14)*

U+2A2C

## Divide (Polar)



*Divide (Polar) —; Niladic: the polar one, 2×1 [1; 0] (new 2026-10-14)*

U+2A38

## Factorial

!

*Factorial —; Niladic ! yields 1 (new 2026-10-12)*

U+0021

## Percent

%

*Percent —; Niladic % is 0 (new 2026-10-12*

U+0025

## Square



*Square — ⊠ X is X squared.; Niladic ⊠ yields 1, derived from Multiply (new 2026-10-12)*

U+22A0

## Multiply



*Multiply —; Niladic ⨯ yields empty array. Replaces Star * U+002A (new 2026-10-12; boxed forms 2026-10-12*

U+2A2F

# Arithmetic: Monadic

## Signum

+

*Ambivalent: monadic Signum (less than zero yields −1, greater than zero yields 1, zero yields 0; back-rank scalar unary)*

U+002B

## Negate

-

*Negate (monadic - X, unary minus)*

U+002D

## Conjugate

⊶

*Conjugate — monadic ⊶X: complex conjugate; negates the imaginary part, preserving orientation (on [r,i] yields [r,−i]; on [r;i] yields [r;−i]) (new 2026-10-15)*

U+22B6

## Double

+2

*Double — monadic +⍨X yields 2 × X (new 2026-10-15)*

U+2A27

## Sum

∑

*Sum — sugar for reduce-addition: ∑ X means / + X (redefined 2026-10-12)*

U+2211

## Square Root

√

*Square Root — monadic √ X yields the square root of X (new 2026-10-15)*

U+221A

## Plus (Complex)

⨁

*Plus (Complex), aka Sum Complex, Add Complex —; Monadic: if X is scalar, sugar for ℙ X 0 (promote to complex); else identity, yields X.*

U+2A22

## Times (Complex)

⨂

*Times (Complex) —; Monadic: if X is scalar, sugar for ℙ X 0 (promote to complex); else identity, yields X.*

U+2A30

## Minus (Complex)



*Minus (Complex) —; Monadic: if X is scalar, sugar for ℚ X 0 (promote to complex); else identity, yields X.*

U+235B

## Times (Polar)



*Times (Polar) —; Monadic: if X is scalar, promote to polar 2×1; else identity, yields X.*

U+2A02

## Divide (Complex)



*Divide (Complex) —; Monadic: if X is scalar, sugar for ℚ X 0 (promote to complex); else identity, yields X.*

U+2A2C

## Divide (Polar)



*Divide (Polar) —; Monadic: if X is scalar, promote to polar 2×1; else identity, yields X.*

U+2A38

## Percent



*Percent —; monadic % X is absolute value, a back-rank operator.; monadic redefined 2026-10-12)*

U+0025

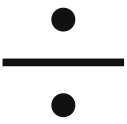
## Star



*Star — exponentiate: monadic * X is 1, because the missing exponent Y is assumed 0, giving X^0 (new 2026-10-12). Idiom: X-th root of Y is * Y ÷ 1 X*

U+002A

## Quotient



*Quotient —; Monadic ÷ X: if X is a square array, yields the matrix inverse; else reciprocal, a back-rank scalar operator (new 2026-10-12)*

U+00F7

## Multiply



*Multiply —; Monadic × X yields X when X numeric; on boxed X, catenates once.; monadic redefined 2026-10-12)*

U+2A2F

## Ceiling



*Ceiling —; monadic ⌈X rounds up*
*(from APL; new 2026-10-12)*

U+2308

## Floor



*Floor —; monadic ⌊X rounds down*
*(from APL; new 2026-10-12)*

U+230A

## Sine



*Sine (new 2026-10-12)*

U+25DD

## Cosine



*Cosine (new 2026-10-12)*

U+25DE

## Arccosine



*Arccosine (new 2026-10-12)*

U+25F6

## Arcsine



*Arcsine (new 2026-10-12)*

U+25F7

## Tangent



*Tangent (new 2026-10-12)*

U+2AD7

## Arctangent



*Arctangent (new 2026-10-12)*

U+2AD8

## Hyperbolic Sine



*Hyperbolic Sine (new 2026-10-12)*

U+2AA6

## Hyperbolic Cosine



*Hyperbolic Cosine (new 2026-10-12)*

U+2AA7

## Hyperbolic Arcsine



*Hyperbolic Arcsine (new 2026-10-12)*

U+2ABD

## Hyperbolic Tangent



*Hyperbolic Tangent (new 2026-10-12)*

U+2AD1

## Natural Log



*Natural Log —; monadic ⍁ X is ln X.; Monadic ⍁ on a polar 2×1 yields the rectangular complex pair [ln r, θ].*

U+237C

## Antilog



*Antilog — sugar for Star: ⍋̇ X is e raised to the X. Monadic ⍋̇ on a rectangular complex pair yields the polar 2×1 [e^a; b] (new 2026-10-12; glyph was 𝒆 U+1D58A until 2026-10-14)*

U+29CA

## Hyperbolic Arctangent



*Hyperbolic Arctangent (new 2026-10-12)*

U+2A4D

## Hyperbolic Arccosine



*Hyperbolic Arccosine (new 2026-10-12)*

U+2ABE

## Factorial

!

*Factorial —; monadic ! X is X factorial*

U+0021

## Fourier



*Fourier — monadic F X: FFT of X (new 2026-10-15)*

U+03DC

## InverseFourier



*InverseFourier — monadic ⅁ X: inverse FFT of X (new 2026-10-15)*

U+2132

## Base Ten Log



*Base Ten Log — ⍙ X is log base 10 of X, sugar for ⍦ 10 X (new 2026-10-12)*

U+2A5F

# Magnitude



*Magnitude — monadic ⍫X: absolute value of X (new 2026-10-16)*

U+2357

# Arithmetic: Dyadic

## Plus



*Ambivalent: dyadic + X Y yields X + Y (addition).; dyadic 2026-10-15)*

U+002B

## Subtract



*Subtract (dyadic - X Y yields X - Y, first minus second), the subtract that works with the / right-fold Reduce.; dyadic 2026-10-12)*

U+002D

## Plus (Complex)



*Plus (Complex), aka Sum Complex, Add Complex —; dyadic: sum of complex numbers (1×2 rectangular), per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before adding.*

U+2A22

## Times (Complex)



*Times (Complex) —; dyadic: product of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before multiplying.*

U+2A30

## Minus (Complex)



*Minus (Complex) —; dyadic: difference of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before subtracting.*

U+235B

## Times (Polar)



*Times (Polar) —; dyadic: product of polar numbers; uses the 2×1 oriented vector to hold magnitude and direction: magnitudes multiply, angles add; if either or both X and Y are scalar, promote to polar 2×1 before multiplying.*

U+2A02

## Divide (Complex)



*Divide (Complex) —; dyadic: quotient of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before dividing.*

U+2A2C

## Divide (Polar)



*Divide (Polar) —; dyadic: quotient of polar numbers; uses the 2×1 oriented vector to hold magnitude and direction: magnitudes divide, angles subtract; if either or both X and Y are scalar, promote to polar 2×1 before dividing.*

U+2A38

## Percent



*Percent —; dyadic % X Y yields the remainder of X divided by Y*

U+0025

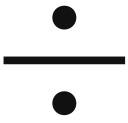
## Star



*Star — exponentiate: dyadic * X Y is X to the power Y (left=base, right=exponent)*

U+002A

## Quotient



*Quotient —; dyadic ÷ X Y divides X by Y, yielding a real; division by zero yields infinity, except 0/0 which yields Nil.*

U+00F7

## Multiply



*Multiply —; dyadic multiplies its arguments; when X is boxed, ⍪X Y catenates X repeated Y times.*

U+2A2F

## Factorial

!

*Factorial —; dyadic ! X Y is the binomial coefficient Y choose X.*

U+0021

## Ceiling

⌈

*Ceiling —; dyadic ⌈ X Y is maximum*

U+2308

## Floor

⌊

*Floor —; dyadic ⌊ X Y is minimum*

U+230A

## Natural Log



*Natural Log —; Dyadic ⍟ X Y is log base X of Y, sugar for ÷ ⍟ Y ⍟ X (new 2026-10-12)*

U+237C

# Square



*Square — ⊠ X is X squared.; Dyadic ⊠ X Y squares X Y times.*

U+22A0

# Constants and Chance: Niladic

## Half

½

*Half —; niladic: 0*

U+00BD

## Pi

π

*Pi — niladic π: value of pi (new 2026-10-14)*

U+03C0

## Bell

𝔅

*Bell —; niladic: random point in (-1, 1), exclusive*

U+1D56D

## Real

ℜ

*Real — niladic: random real number between 0 and 1; sugar for ½ + 1 𝔅 (new 2026-10-13)*

U+1D57D

## Gamma



*Gamma — niladic γ: Euler-Mascheroni constant (new 2026-10-14)*

U+03B3

## Golden



*Golden — niladic φ: golden mean (new 2026-10-14)*

U+03C6

## Euler



*Euler — niladic ℇ: Euler's constant (new 2026-10-14)*

U+2107

## Dice



*Dice —; niladic: single 6-sided die (1-6)*

U+1D56F

## D2



*D2 — niladic: a single 2-sided die (1-2), the die face U+2681 (2026-10-20)*

U+2681

## D4



*D4 — niladic: a single 4-sided die (1-4), the die face U+2683 (2026-10-20)*

U+2683

## D6



*D6 — niladic: a single 6-sided die (1-6), the die face U+2685 (2026-10-20)*

U+2685

## Pick



*Pick —; niladic: Nil*

U+2A03

# Constants and Chance: Monadic

## Half

½

*Half —; monadic ½ X: X divided by 2; array → sugar for ⊞(ρ X); train → floor(length/2) index; dictionary → Nil (new 2026-10-13)*

U+00BD

## Pick



*Pick —; monadic ⍁X: scalar → same scalar; train → use Dice of size train to select element; array → use Dice of sides from shape to select element; dictionary → random {key:value} pair (new 2026-10-13)*

U+2A03

## Bell

𝔅

*Bell —; monadic 𝔅μ: bell curve point with mean μ (μ in (-1,1))*

U+1D56D

## Dice

𝔇

*Dice —; monadic 𝔇X: single X-sided die (1-X)*

U+1D56F

# Deal



*Deal — given a scalar, works as expected; given an array, fills the entire back rank with limited values. Concrete examples and summary TBD (new 2026-10-11)*

U+003F

# Constants and Chance: Dyadic

## Bell

𝔅

*Bell —; dyadic 𝔅 μ σ: bell curve point with mean μ and stddev σ (σ ≤ 1/3) (new 2026-10-13)*

U+1D56D

## Dice

𝔇

*Dice —; dyadic 𝔇 X N: N X-sided dice; all sugar via ℜ (hence 𝔅) (new 2026-10-13; die face U+2685 tried and reverted 2026-10-20)*

U+1D56F

# Number Theory: Monadic

## Divisible



*Divisible —; monadic ⍱ X yields Truth when X is divisible by 2, Nil otherwise*

U+23A2

## Indivisible



*Indivisible —; monadic ⍲ X: sugar for ¬ ⍱ X (the Odd test)*

U+2AEE

## Primes



*Primes — monadic 𝒫X yields the prime factors of X, per J's q:. (new 2026-10-18)*

U+1D4AB

## Prime



*Prime — monadic 𝔓 N yields the Nth prime, per J's p: (zero-indexed: 𝔓 0 is 2). (new 2026-10-18)*

U+1D513

## Prime



*Prime — monadic ⌽ X: sugar for ⍵/⍨ X ⍳ 2 ⌊ √ X ⍝ ≤ X 1 (new 2026-10-15)*

U+23C0

## Factor



*Factor — factorization verb (new 2026-10-16)*

U+27D2

## Combinatoric



*Combinatoric —; monadic ⍗ X: factorial of X*

U+2A69

# Number Theory: Dyadic

## LCM



*LCM — dyadic ⦶ X Y: least common multiple of X and Y (new 2026-10-16)*

U+238A

## Divisible



*Divisible —; dyadic ⁄ X Y yields Truth when X is divisible by Y, Nil otherwise (new 2026-10-15)*

U+23A2

## GCD



*GCD — dyadic ⫻ X Y: greatest common divisor of X and Y (new 2026-10-16)*

U+2980

## Indivisible



*Indivisible —; dyadic ⫲ X Y: sugar for ¬ ⁄ X Y (boolean Not of Divisible) (new 2026-10-15)*

U+2AEE

# Combinatoric



*Combinatoric —; dyadic ⋕ X Y: binomial X choose Y (new 2026-10-16)*

U+2A69

# Polynomials and Roots: Monadic

## NextLocal



*NextLocal — the Roots division step: given P, yields [R prime, P prime] where R prime is Local(P) and P prime is P deflated by synthetic division of R prime. (new 2026-10-19)*

U+2113

## Roots



*Roots — sugar: ℞ P is ℜ ℓ P (Reveal NextLocal over P); yields all polynomial roots. (new 2026-10-18; factored via Reveal 2026-10-19)*

U+211E

## Differentiate



*Differentiate — train verb ∂ T yields symbolically differentiated version of train T of coefficients, multiplying constants as needed; same order as Polynomial (new 2026-10-13)*

U+2202

## Integrate



*Integrate — train verb ∫ T yields symbolically integrated version of train T of coefficients, dividing constants as needed; same order as Polynomial (new 2026-10-13)*

U+222B

## Solve

&

*Solve — monadic & A: array → vector of polynomial coefficients; missing input row is all zeroes (new 2026-10-13)*

U+0026

## Optimize

𝝯

*Optimize —; monadic 𝝯 V: vector → polynomial; searches free variable with delta = 0.01 * domain, epsilon defaults to 1e-6*

U+214B

## Local

𝔏

*Local — monadic ⍵ V is 𝝯 ⫴𝔏 & V (FixedPoint Optimize Triple Solve V); approximates best x for a local minimal of V (new 2026-10-13)*

U+1D50F

# Polynomials and Roots: Dyadic

## Reveal



*Reveal — intrinsic. Dyadic ℜ P Y yields the successive application of P upon Y as Y shrinks to Nil. Epsilon from P has two elements: an alpha to apply P to next Y, and the next Y. (new 2026-10-18)*

U+211C

## Optimize



*Optimize —; dyadic 𝔅 V ε: recurse into smaller domains until change in c < ε (new 2026-10-13)*

U+214B

## Integer Division



*Integer Division — dyadic ⫽ X Y divides X by Y, yielding an integer (replaces Slash / integer division from manual p.19; / is now Reduce). Vector form: ⫽ V X where V is vector yields synthetic quotient of V by X (new 2026-10-14)*

U+2AFD

## Evaluate



*Evaluate — dyadic 𝒇X T evaluates train T of coefficients upon X (new 2026-10-13)*

U+1D58B

# FixedPoint



*Fixed point — dyadic cø V guess: refine guess until epsilon < 1e-6 (max 100 steps); failure to converge returns infinity (new 2026-10-13)*

U+AB44

# Matrices: Niladic

## Outer Product



*Outer Product —; Niladic ⊗ yields 1 (new 2026-10-12)*

U+2297

## Inner Product



*Inner Product —; Niladic · yields 1 (new 2026-10-12; matrix case 2026-10-14)*

U+22C5

## Multiply Matrices



*Multiply Matrices —; Niladic ⋈ yields the 1×1 identity matrix (new 2026-10-14)*

U+22C8

# Matrices: Monadic

## Outer Product



*Outer Product —; Monadic ⊗ X yields X.*

U+2297

## Inner Product



*Inner Product —; Monadic ⋅ X yields X.*

U+22C5

## Multiply Matrices



*Multiply Matrices —; Monadic ⋈ X yields X.*

U+22C8

## L Factor



*L Factor — monadic: the L (lower triangular, 1s on diagonal) factor of the LU decomposition of a rank-2 array (new 2026-10-14)*

U+1D577

## Determinant



*Determinant — monadic ⌹X: determinant of square matrix X (new 2026-10-16)*

U+233C

## U Factor



*U Factor — monadic: the U (upper triangular) factor of the LU decomposition of a rank-2 array (new 2026-10-14)*

U+1D580

# Matrices: Dyadic

## Outer Product



*Outer Product —; dyadic ⊗ X Y yields the outer product matrix where entry [i,j] is ⍺ X[i] Y[j]; shape is the concatenation of the shapes of X and Y.*

U+2297

## Inner Product



*Inner Product —; dyadic ⋅ X Y: for vectors X, Y, the dot product (sum of pairwise products); for matrices, the Frobenius inner product: a scalar, the sum of all pairwise products.*

U+22C5

## Multiply Matrices



*Multiply Matrices —; dyadic ⋈ X Y: matrix multiplication, (⋈ X Y)[i;j] = ⋅ X[i;] ( ⌽ Y)[j;]; inner dimensions must agree, non-square fine.*

U+22C8

# Arrays: Niladic

Pee

ℙ

*Pee — Where⊡Pee yields Where [0, 1 (first two coordinates)*

U+2119

Zee

ℤ

*Zee — the last position of a slice (Where⊡Zee yields Where [-1)*

U+2124

Exx

𝕏

*Exx — the zero position of a slice (Where [0)*

U+1D54F

Wye

𝕐

*Wye — the midposition of a slice*

U+1D550

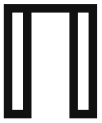
## Extend



*Extend —; niladic ⊢: sugar for ⊂ Nil*

U+213E

## Catenate



*Catenate —; niladic ⊣: sugar for ⊂ Nil*

U+213F

## Grade Up



*Grade Up — ⍋X yields the indices that sort X ascending. Niladic ⍋ yields 0 (from APL; new 2026-10-12)*

U+234B

## Grade Down



*Grade Down — ⍒X yields the indices that sort X descending. Niladic ⍒ yields Nil (from APL; new 2026-10-12)*

U+2352

# Arrays: Monadic

## Rank



*Rank — monadic ≠ X yields the number of dimensions in X; zero when scalar. (new 2026-10-18)*

U+22D5

## Permute



*Permute —; monadic ⍨ X and dyadic X ⍨ Y work as in J's C.. Both modes intrinsic. (new 2026-10-19)*

U+2A0A

## Laminate



*Laminate —; monadic ⍪ X: adds a leading singleton axis*

U+2A5E

## Anagram



*Anagram — permutation index/parity (J's A.). Both modes intrinsic. (new 2026-10-19)*

U+1D49C

## Enumerate

№

*Enumerate — monadic № X: sugar for ⍕⍳⍴X; yields index-value pairs for X (new 2026-10-18)*

U+2116

## Reflect

⌿

*Reflect — monadic ⌿X: sugar for ⍕⌽-⌽⍳⍴X (new 2026-10-15)*

U+223B

## Count

76

⍳

U+2373

## Kount

⍭

*Kount — monadic ⍭B: takes a vector of T or Nil; returns a vector counting the T, turning each Nil to 0. Intrinsic. (new 2026-10-18)*

U+2A23

# ⍳ Count

U+2373

Count — magical: when given a shape, fills an array matching shape. Scalar: behaves as normal. Array: fills array with next dimension limited by scalar from back rank — e.g. ⍳ 2 3 → rows of 0 1 2 (origin 0; last axis counts, outer axes replicate) (new 2026-10-11)

## RavelRows



*RavelRows — monadic ⍪X: yields the rows of X as a vector (Antitable; new 2026-10-15)*

U+22F2

## Ravel



*Ravel — monadic ⍎X: flattens array X to vector, row-major (new 2026-10-14)*

U+22F3

## Reverse



*Reverse —; monadic ⌽ X reverses elements along the last axis*

U+233D

## Transpose



*Transpose — monadic ⍉ X reverses the order of axes (from APL; new 2026-10-12)*

U+2349

## Shape



*Shape —; monadic ρ X: shape of X; scalar → 0; array → vector of dimensions; train → [1, length of train]; dictionary → [1, pairs present]*

U+03C1

## Extend



*Extend —; monadic ⍮ X: Table X as in APL, adds trailing axis of length 1*

U+213E

## Catenate



*Catenate —; monadic ⍪ X: catenates X to itself*

U+213F

## Rotate



*Rotate —; monadic ⌽ X: rotate X toward origin by 1*

U+29BD

## Cardinality



*Cardinality — "how big is this?"; scalar → -1; array → number of elements held; dictionary → number of pairs; train → length of train (new 2026-10-13)*

U+2210

## Center



*Center — takes a shape, returns its integer halves expressed as a vector (e.g. [10, 6] → [5, 3]) (new 2026-10-13)*

U+229E

## Sight



*Sight — takes 2D array, finds center, returns vector location of nearest non-zero scalar; ties pick one (new 2026-10-13)*

U+2388

# Arrays: Dyadic

## Put



*Put — ⫤⊨ X P A: value at coordinate P of array A becomes X (new 2026-10-18)*

U+27DA

## Permute



*Permute —*

U+2A0A

## Laminate



*Laminate —; dyadic ⩢ A B: yields [A; B]; when A and B are both scalars, yields A B (new 2026-10-17)*

U+2A5E

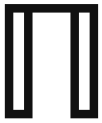
## Anagram



*Anagram — permutation index/parity (J's A.). Both modes intrinsic. (new 2026-10-19)*

U+1D49C

## Catenate



*Catenate —; dyadic ⎍ X Y: catenate X and Y as in APL; variadic ⎍ A ... Z: catenate A through Z left to right (new 2026-10-14)*

U+213F

## Take



*Take — ↑ X Y takes the first X elements of Y (last X if X is negative); X may be a vector for per-axis counts (from APL; new 2026-10-12)*

U+2191

## Drop



*Drop — ↓ X Y drops the first X elements of Y (last X if X is negative); X may be a vector for per-axis counts (from APL; new 2026-10-12)*

U+2193

## Reverse



*Reverse —; dyadic ⌽ X Y rotates Y by X positions (from APL; new 2026-10-12)*

U+233D

## Shape

ρ

*Shape —; dyadic ρ S X: reshape X to shape S (per APL) (new 2026-10-13)*

U+03C1

## Extend



*Extend —; dyadic ⍋ X N: extends by N axes (new 2026-10-14)*

U+213E

## Rotate



*Rotate —; dyadic ⍜ X Y: rotate X by Y (Y positive toward origin, negative away) (new 2026-10-16)*

U+29BD

# Boxes: Niladic

## Box



*Box —; Niladic ⊂ yields Truth*
*(from APL Enclose; new 2026-10-12)*

U+2282

## Unbox



*Unbox —; Niladic ⊃ yields Nil*
*(from APL Disclose; new 2026-10-12)*

U+2283

## Summarize



*Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar; niladic ⊚ is the 1,1 vector containing zero.*

U+229A

# Boxes: Monadic

## Box



*Box — monadic ⍈ A: wraps A in a box so it travels as a single element (new 2026-10-17)*

U+2256

## Unbox



*Unbox — monadic ⍇ A: opens the box, yielding its contents (new 2026-10-17)*

U+2257

## Catalogue



*Catalogue —; monadic ⍀ X: cartesian product of the boxed lists in vector X (new 2026-10-17)*

U+2A37

## Raze



*Raze — monadic ⫻ A: sugar for Ravel Map Unbox A; opens every box and joins the contents into one vector (new 2026-10-17)*

U+2AF5

## Box



*Box —; monadic ⊂X encloses X into a scalar box.*

U+2282

## Unbox



*Unbox —; monadic ⊃X yields X.*

U+2283

## Summarize



*Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar*

U+229A

## Zip

90



U+22B0



# Zip

`U+22B0`

Zip — takes a train of vectors and zips them: ⍮ V1 V2 ... Vn yields an m×n matrix (m = vector length, n = train length); the resulting width is the length of the train; a scalar in the train expands into a column to fit the others; a container in the train contributes only its origin scalar, recursively (the summary effect); mismatched lengths Nil-pad (consistent with Diction); zips along the back rank for higher-rank arrays; ⍮ 1 2 yields the column [1;2] (new 2026-10-12; recycles the retired AntiSignum glyph)

# Boxes: Dyadic

## Summarize

93



U+229A

## Fetch



*Fetch — dyadic ⊸V X: applies V as search vector into X's items; follows path V down through nested boxes (new 2026-10-17)*

U+22B8

## Catalogue



*Catalogue —; dyadic ⊗A B: yields [[A0,B0] ... [A1,B1]] ... [[AN,B0] ... [AN,BN]], the outer product matrix of pairs*

U+2A37

## Link



*Link — dyadic ⫟X Y: sugar for Catenate Map Box X Y; boxes X and Y into a two-element list (new 2026-10-17)*

U+2AE9



# Summarize

U+229A

Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar; Dyadic D ⊚ A: D (Direction) is a rank-1 array of coordinates; summarization starts at A[D], then proceeds as monadic; a scalar Direction s means start at (s, 0); Zip summarizes each train element (new 2026-10-12)

## Box



*Box —; Dyadic ⊂ X Y encloses X by Y levels; Y=0 leaves X unchanged.*

U+2282

## Unbox

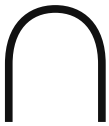


*Unbox —; Dyadic ⊃ X Y picks element X from Y.*

U+2283

# Sets and Membership: Niladic

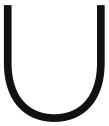
## Intersection



*Intersection —; niladic ∩: sugar for ⊂ Nil*

U+2229

## Union

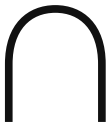


*Union —; niladic ∪: sugar for ⊂ Nil*

U+222A

# Sets and Membership: Monadic

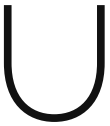
## Intersection



*Intersection —; monadic ∩ A: is A*

U+2229

## Union



*Union —; monadic ∪X: unique as in APL*

U+222A

## InOf



*InOf — the curry of In: ⇦InOf ⍀∈ X; monadic predicate testing membership in X. (new 2026-10-18)*

U+22F8

## Where



*Where — monadic ⍸Y: returns vector of indices of all elements of array Y that are Truth (new 2026-10-18)*

U+2A42

# Enlist

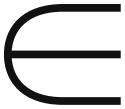


*Enlist —; monadic ∈Y: as defined in APL (deep flatten to vector)*

U+220A

# Sets and Membership: Dyadic

## In



*In — dyadic ∈ Y X yields Truth when X is in Y, Nil otherwise (J's e.). (new 2026-10-18)*

U+2208

## Bucket



*Bucket — dyadic ∐ X Y: yields index of interval in X that Y falls into (new 2026-10-18)*

U+2210

## InLast



*InLast — dyadic ∈̇ X Y: yields the index of the last occurrence of X in Y; yields length of Y if not found (new 2026-10-18)*

U+22F5

## InAt



*InAt — dyadic ⋲ X Y: yields the index of Y whose value equals X; yields length of Y if not found (new 2026-10-18)*

U+22F9

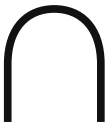
## In



*Enlist —; dyadic ∈ X Y: find/membership as defined in APL (new 2026-10-13)*

U+220A

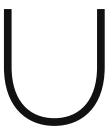
## Intersection



*Intersection —; dyadic ∩ A B: as in APL, elements in both (new 2026-10-14)*

U+2229

## Union



*Union —; dyadic ∪ X Y: union as in APL (new 2026-10-14)*

U+222A

## Set Difference



*Set Difference — dyadic ⍠A B: elements in A but not in B (new 2026-10-14)*

U+2A43

# Trains: Niladic

## Reduce



*Reduce — / F V: start at the highest index (its scalar is the initial result), step down toward origin; at each step result = F(scalar, result). Right fold.; Niladic / returns 0 (new 2026-10-12)*

U+002F

## Map



*Map — ⍬ A B executes function A with zero arguments upon all scalars of B, as a back rank operator; usual back scalar rules.; Niladic ⍬ yields Truth, same as niladic 𝔗 (new 2026-10-12)*

U+226C

## Reduce (forward)



*⌿ F V: start at the origin (its scalar is the initial result), step up toward the highest index; at each step result = F(result, scalar). Left fold, the reversed scan.*

U+233F

## Identity



*Identity — 𝔗 V yields V unchanged; on a train, yields the train unchanged. Niladic 𝔗 yields Truth (new 2026-10-12)*

U+1D574

## Backstroke



*Backstroke — Scan: the running Reduce, paired with /. \ F V on a length-n V yields a length-n vector where element i is / F (right fold) applied to the first i elements of V.; Niladic \ yields 0 (new 2026-10-12)*

U+005C

## Stop



*Stop — statement separator; separates statements on one line*

U+22C4

## Backstroke (forward)



*Backstroke (forward) — Reverse Scan, paired with ⍀. ⍀ F V on a length-n V yields a length-n vector where element i is ⍀F (left fold) applied to the first i elements of V.*

U+2340

## Strip



*Strip —; niladic: removes the train from head up to but not including the first Stop encountered*

U+2ACF

# Trains: Monadic

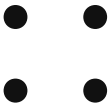
## Meter

110



U+2214

## Atomize



*Atomize — monadic ∴A: reads up to the first 32 elements of A (origin to end) and prepares them for use by train elements ahead (new 2026-10-14)*

U+2237

## Duplicate



*Duplicate — copies a train to a train twice as long: ⊷A B C → A B C A B C (was ==U+2A75, then ∶ U+2236 — settled 2026-10-12)*

U+22B7

## Caboose



*Caboose — picks up the train from head to Stop, then converts boxed contents to a vector (inverse of Atomize) (new 2026-10-15)*

U+2A78



# Meter

U+2214

Meter — measures its argument: ∔ V is ≺ + ⊷ V (zip the needle with the signal); yields M, a sequence of (needle, signal) pairs, needle ∈ -/0/+. M works in place of V in all Collate modes: the needle selects the function, the function transforms the signal (was ±, corrected to ≺ 2026-10-12)

## Reduce



*Reduce — / F V: start at the highest index (its scalar is the initial result), step down toward origin; at each step result = F(scalar, result). Right fold.; Monadic / V yields a linear array of the scalars ripped from V, highest to lowest indices.*

U+002F

## Map



*Map — ⍁ A B executes function A with zero arguments upon all scalars of B, as a back rank operator; usual back scalar rules.; Monadic ⍁ B is identical to Identity 𝔗 B.*

U+226C

## Reduce (forward)



*⌿ F V: start at the origin (its scalar is the initial result), step up toward the highest index; at each step result = F(result, scalar). Left fold, the reversed scan.; Monadic ⌿ V and niladic ⌿ behave identically to / (new 2026-10-12)*

U+233F

## Zero



*Zero — on a train, replaces all values with zeroes while keeping shapes (new 2026-10-12)*

U+1D585

## Backstroke

113

\

U+005C

## Duplicate Head

Ⅱ

*Duplicate Head — Ⅱ A ... yields A A ...; duplicates head of train, aka duplicate/dup (new 2026-10-14)*

U+2161

## Backstroke (forward)

⍀

*Backstroke (forward) — Reverse Scan, paired with ⌿. ⍀ F V on a length-n V yields a length-n vector where element i is ⌿F (left fold) applied to the first i elements of V.; Monadic ⍀ V and niladic ⍀ behave identically to \ (new 2026-10-12)*

U+2340

## Pipeline

⫷

*Pipeline —*

U+2AF7



# Backstroke

`U+005C`

Backstroke — Scan: the running Reduce, paired with /. \ F V on a length-n V yields a length-n vector where element i is / F (right fold) applied to the first i elements of V.; Monadic \ V yields a linear array of the scalars ripped from V, lowest to highest indices.

## Head



*Head — sugar for ⊐1 (new 2026-10-15)*

U+2991

## Tail



*Tail — sugar for ⊐-2 (new 2026-10-15)*

U+2992

## Duplicate Tail



*Duplicate Tail — ⧟... A yields ... A A; duplicates tail of train (new 2026-10-13)*

U+29DE

## Strip

115



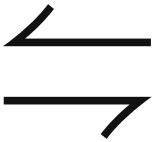
U+2ACF

# ⊐ Strip

U+2ACF

Strip —; monadic ⊐ X: removes the train from position X (0 is head) up to but not including the first Stop encountered; when X < 0, strips from position X in reverse order back to head, where −1 is the final tail position adjacent to the first Stop encountered (new 2026-10-15)

## Swap



*Swap — ⇋A B ... yields B A ...; swaps head two of train. Single: ⇋ A Stop X yields X. Bare: ⇋ Stop X yields X (new 2026-10-13)*

U+21CB

## Replicate

117



U+232C

## Rotate Head



*Rotate Head — ⟲A B C ... yields B C ... A; rotates head to tail (new 2026-10-13)*

U+27F2

## Rotate Tail



*Rotate Tail — ⟳ ... A B C yields C ... A B; rotates tail to head (new 2026-10-13)*

U+27F3



# Replicate

U+232C

Replicate — ⌬ N X Stop yields X replicated N times; e.g. ⌬ 5 A Stop gives A A A A A. Zero case: ⌬ 0 A Stop X yields X. Negative N is antireplicate: ⌬ -1 A A A A A Stop yields A A A A. Empty antireplicate: ⌬ -1 Stop X yields X. Bare: ⌬ Stop X yields X (new 2026-10-13)

## Nip



*Nip — ↢ A B ... yields A ...; drops second, keeps head (new 2026-10-13)*

U+21F7

## Tuck



*Tuck — ↣ A B ... yields A B A ...; copies head under second (new 2026-10-13)*

U+21F8

## Reverse Train



*Reverse Train — reverses order of train (new 2026-10-13)*

U+21F9

## Over



*Over — ⤺ A B ... yields B A B ...; copies second to head (new 2026-10-13)*

U+293A

## Triple

Ⅲ

*Triple — Ⅲ X yields X X X; triplicates top of stack (new 2026-10-13)*

U+2162

## Fork



*Fork — ⋔ F G H X: sugar for G H Swap F Dup X (J fork; new 2026-10-14)*

U+22D4

## Drop Train



*Drop Train — sugar for reverse, replicate -1, reverse; drops head of train (new 2026-10-13)*

U+291A

# Trains: Dyadic

## Applyself



*Applyself — ⍣⍤1 f n a applies f n times to a; sugar for ⫷a ⍵⊂⍣⍺⍵ f n (from APL power; new 2026-10-12)*

U+2363

## Pipeline



*Pipeline —; dyadic ⫷X Y applies vector Y, readable as a train of monadic functions, in sequence to X (new 2026-10-12)*

U+2AF7

# Functions: Monadic

## Diction



*Diction — dict-backed decision-table constructor (was Vectorize, renamed 2026-10-11). ¡ A B → {A:B}; ¡ A B C D → {A:B,C:D}; monadic ¡ (a b c) self-maps into a set. Membership sugar: X member Y → ⊤(X ¿ (¡ Y))*

U+00A1

## Glyph



*Glyph — the name for a single character value. Glyph ← ≡ ⌧: Glyph's function is the literalize train (via monadic ≡) — to make a Glyph, literalize*

U+00AE

## Decide

125



U+00BF

## Function



*Function — gets glyph del (∇, nabla): the function-definition operator. Monadic ∇Y executes array Y as an FSA. (monadic 2026-10-19)*

U+2207

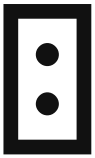


# Decide

U+00BF

Decide — first-match generalized switch/match over n×2k train tables (was When, 2026-10-11). n×2 returns 1×1 (scalar); n×4 returns 1×2; generally n×2k returns 1×k. An explicit trailing C is unmatched-result filler, not an else branch; without filler, unmatched returns Nil without logging an error. Canonical: R ← X ¿ ¡ A B C

## Print



*Print — the quad-colon verb: prints the value from its right, then passes that value to its left (new 2026-10-11)*

U+2360

## (literalize)



*Literalize next nonspace char, or ※0x???? by codepoint*

U+2A33

## Rail



*Rail — monadic ⌐¯X yields a vector of symbols from train X (train-to-array conversion). (new 2026-10-19)*

U+2ACD

## Lay



*Lay — monadic ¯¬V yields the train laid down from origin to limit of vector V (array-to-train, inverse of Rail). (new 2026-10-19)*

U+2ACE

# Lambda

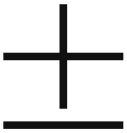


*Lambda — lambda abstraction (new 2026-10-13)*

U+039B

# Functions: Dyadic

## Collate



*Collate — executes one of three trains upon a value: for less than zero, for zero, or for greater than zero (was AntiSignum ≺ U+22B0, renamed 2026-10-12; complement to signum)*

U+00B1

## Curry

130



U+1D49E



# Curry

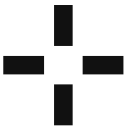
U+1D49E

Curry — intrinsic. Partial application: turns a dyadic verb into a monadic predicate by binding an argument, e.g. making = X from =. Semantics: Gets Q Curry P args: args to omega prime, omega prime shifts into alpha prime (binding the leftmost argument, textbook curry order), Q X makes X alpha, gets slice -1 of omega prime A, then call P; Q alpha yields P evaluated upon alpha prime, omega prime, and alpha as arguments in that order. Omega becomes nil upon Q's execution. Principle: Gets Q Curry P train is the principle for creating curries (Q is an example name assigned to the curried function, not a verb). (new 2026-10-18)

# Errors: Niladic

## Silence

134



U+22B9

## Compress

135



U+22D6

## Errline



*Last error: which line*

U+274C

## Errmsg



*Last error: what (type)*

U+274E



# Silence

U+22B9

Silence — ⊹ Z V drops all nonzero values; Z transforms the zero contents in sequence. ⊹ Z M on metered M: a nonzero needle drops the pair; a zero needle selects Z (applied to the signal).; Niladic ⊹ yields boxed Nil, a trial vector for enclosing operators to work on for starts (the dual of Compress; new 2026-10-12)



# Compress

U+22D6

Compress — ⋖ P V applies transform P to each nonzero content of V; zero contents are eliminated; sequence order preserved. ⋖ P M on metered M: a nonzero needle selects P (applied to the signal); a zero needle drops the pair.; Niladic ⋖ yields boxed Truth, a trial vector for enclosing operators to work on for starts (a comprehension filter; new 2026-10-12)

## Cut

*Cut —; niladic: on error, erases the error and returns Nil*

U+22EE

## Debugger

✔⋄ *enters debugger iff error registers set*

U+2714

## Errdmd

*Last error: diamond position in line*

U+2727

## Succeed

*Succeed — like Cut, except returns Truth (new 2026-10-15)*

U+2999

# Anticut



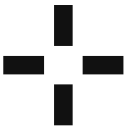
*Anticut — on error throws to debugger; on return from debugger, returns Nil*

U+22EF

# Errors: Monadic

## Silence

140



U+22B9

## Compress

141



U+22D6

## Cut



*Cut —; monadic ⁝ X: checks X for Truth before returning*

U+22EE



# Silence

U+22B9

Silence — ⊹ Z V drops all nonzero values; Z transforms the zero contents in sequence. ⊹ Z M on metered M: a nonzero needle drops the pair; a zero needle selects Z (applied to the signal).; Monadic ⊹ V or ⊹ M means Z is identity: keep zeros (or zero-needle signals) as-is, drop nonzeros.



# Compress

U+22D6

Compress — ⋖ P V applies transform P to each nonzero content of V; zero contents are eliminated; sequence order preserved. ⋖ P M on metered M: a nonzero needle selects P (applied to the signal); a zero needle drops the pair.; Monadic ⋖ V or ⋖ M means P is identity: pure filter, no transform.

# Classes and Properties: Niladic

## Constant

145

α

U+03B1

## Epsilon

ε

*Epsilon — holds the result of any getter, setter, or nabla when no explicit return glyph is used*

U+03B5

## Omega

ω

*Omega — the vector of arguments passed to a setter (◅)*

U+03C9

# α Constant

`U+03B1`

Constant — declares a class constant: X α name ← value names a constant for class X. α is also the original (pre-override) value: read-only, may not be changed directly; trying registers an error. As the original value α is an ad hoc constant — contextual to the setter, not a declared class constant

# Classes and Properties: Monadic

## Increment



*Increment —; monadic ⊕ D: invokes D's setter with nabla + 1 as the xthrement.*

U+2295

## Setter



*Setter — the setter nabla: defines a property setter, as ∇ defines a function. Generalized: takes an xthrement argument — modifies instance data (increment/decrement generalized) rather than assigning (new 2026-10-14)*

U+22B2

## Getter



*Getter — the getter nabla: defines a property getter, as ∇ defines a function*

U+22B3

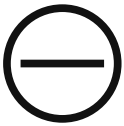
## Property



*Property —; Monadic ⋘ D: yields the result of D's getter (getter mode of D's class call) (new 2026-10-14)*

U+22D8

## Decrement



*Decrement —; monadic ⊖ D: invokes D's setter with nabla - 1 as the xthrement.*

U+2296

## Divide By



*Divide By —; monadic ⊘ D: invokes D's setter with nabla ÷ 2 as the xthrement (halve).*

U+2298

## Multiply By



*Multiply By —; monadic ⊛ D: invokes D's setter with nabla × 2 as the xthrement (double).*

U+229B

# Classes and Properties: Dyadic

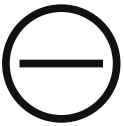
## Increment



*Increment —; Dyadic ⊕ D X: invokes D's setter with nabla + X as the xthrement (new 2026-10-14)*

U+2295

## Decrement



*Decrement —; Dyadic ⊖ D X: invokes D's setter with nabla - X as the xthrement (new 2026-10-14)*

U+2296

## Multiply By



*Multiply By —; Dyadic ⊛ D X: invokes D's setter with nabla ⌿ X as the xthrement (new 2026-10-14)*

U+229B

## Property



*Property —; dyadic ⋘ I D: invokes D's getter (getter mode of D's class call), then invokes I's setter (setter mode of I's class call) with the result as argument; yields the assigned data as a pass-through.*

U+22D8

## Divide By



*Divide By —; Dyadic ⊘ D X: invokes D's setter with nabla ÷ X as the xthrement (new 2026-10-14)*

U+2298

## (method binding)



*Binds method to object: X⊡function*

U+22A1

## Gets



*Gets — ⇔A G: A stored in symbols[G], G stored in instances[A]; bidirectional binding (new 2026-10-13)*

U+2B04

# The Virtual: Niladic

## IntegerBounded



*IntegerBounded — bounded integer class: min/max enforced on creation, assignment, math; clamp + silent sticky flag on violation (new 2026-10-15)*

U+012A

## Trapdoor



*Trapdoor — the assembly trap door: invokes machine code at a given address (installed via ctypes; proven 2026-10-14).*

U+1D21D

## VirtualSpace



*VirtualSpace — the VR coordinate space: navel origin, forward half-sphere, [r;θ;φ], angles ±32767, distance in mm (new 2026-10-15)*

U+1D4E5

## Prim



*Prim — base VR primitive: name, position, collision machinery; posted via Virtual (new 2026-10-16)*

U+1D7D8

Raster



*Raster — VR prim ⊑ Prim: array of pels, rank 4 [Y,X,1,3] (new 2026-10-16)*

U+1D7D9

Point



*Point — VR prim ⊑ Prim: dot prim, [N,6] (H,V,D,RGB) (new 2026-10-16)*

U+1D7DA

Triangle



*Triangle — VR prim ⊑ Prim: 9 points × 2 thetas + D + RGB = 22 numbers, [N,22] (new 2026-10-16)*

U+1D7DB

# The Virtual: Monadic

Trapdoor

158



U+1D21D

# ⊂ Trapdoor

`U+1D21D`

Trapdoor — the assembly trap door: invokes machine code at a given address (installed via ctypes; proven 2026-10-14).; Monadic ⊂ A: puts the address of A into the accumulator before calling. ⇔ X ⊂ ∇ assembly ∇: X gets the address of the compiled assembly script. If X is so bound, niladic X calls the code. ε maps to the accumulator. The call yields ε

# Dungeon

# Dungeon: Places

## Where

@

*Where the hero is — bound to '@', the roguelike player glyph. Where⊡Zee yields Where [-1 (last); Where⊡Wye yields Where [1 (midposition — middle of 3 coords); Where⊡Exx yields Where [0 (zero). INDICES START AT ZERO*

U+0040

## Dungeon

£

*The Dungeon — literally the set of all levels*

U+00A3

## Screen

§

*The Screen — intrinsic 20×60 array (16 play + 4 I/O rows)*

U+00A7

## Level

¶

*Level — the pilcrow: a single dungeon level, member of £ Dungeon*

U+00B6

## Cell

¢

*A dungeon cell — the cent sign joins the family: § £ ¢. Cell⊡location ← £ Where: the cell's location is the Dungeon at Where. A dungeon square IS a dictionary of instances (the ⓘ registry's entries present at that square)*

U+00A2

## Up



*Up — the upward/north movement. Where⊡up yields ∅ when Where⊡Zee is 0, else Where with Zee decremented (refined 2026-10-10: boundary-checked — top edge is a quiet Nil, not a negative index)*

U+25B2

## Down



*Down — the downward/south movement, counterpart to Up. Where⊡down yields ∅ when Where⊡Zee ≥ Dungeon⊡Zee (bottom edge), else Where⊡Zee + 1 (confirmed 2026-10-10)*

U+25BC

# Dungeon: Classes

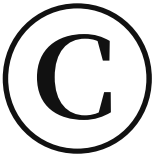
## Class



*Class — the circled dot. X⊡Class gets a RANK-2 ARRAY OF TRAINS: a class IS a decision table made of trains. Class instances are callable; the train defaults to Nil (∅)*

U+2299

## Classes



*Classes — the set of all classes (plural of Class), parallel to £ Dungeon as the set of all levels. Implementation: a single twin (bidirectional word↔glyph) dictionary at global scope*

U+24B8

## Instances

166



U+24BE

## Instance



*Instance — the INDIVIDUAL instance (singular), distinct from Ⓘ the table. Joins the blackboard family (𝕏 𝕐 ℤ ℙ 𝕆). Carries the (glyph, hash-at-creation) identity tuple*

U+1D540



# Instances

U+24BE

Instances — the TABLE/set of all instances (plural), parallel to Ⓒ Classes. Implementation: a single twin (bidirectional word ↔ glyph) dictionary at global scope. Instance identity is the (glyph, hash-at-creation) tuple, used as the key field in both the global instance dictionary and each square (Cell) dictionary

## Interpreter



*Interpreter — the Wuxia interpreter (new 2026-10-15)*

U+203D

## Inherits



*Inherits — the inheritance operator (square subset of)*

U+2291

## Isa



*Isa — the is-a / instantiation operator (2026-10-10). Monadic ⋹ Y: Y is a class to copy data from; new instance.ε = Y.ε. Dyadic X ⋹ Y X: X is a class, optional initializer; new instance.ε = X.ε (new 2026-10-14)*

U+22FF

## Object



*Object — a DUNGEON object, not a generic OOP object. The blackboard O joins the projection family (𝕏 𝕐 ℤ ℙ): the root of the dungeon's object hierarchy, what every dungeon entity inherits from*

U+1D546

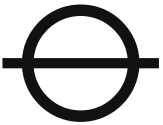
## VirtualDistance



*VirtualDistance — distance in VirtualSpace, 0-512 (new 2026-10-15)*

U+27F0

## Angle



*Angle — a VR angle (theta/phi) (new 2026-10-15)*

U+29B5

## VirtualHorizontal



*VirtualHorizontal — horizontal angle in VirtualSpace (new 2026-10-15)*

U+29D3

## VirtualVertical



*VirtualVertical — vertical angle in VirtualSpace (new 2026-10-15)*

U+29D7

# Dungeon: Objects

## Ring



*Ring — the blackboard R: a dungeon object (ring), third of the dungeon entity glyphs*

U+211D

## Wand



*Wand — left tack: a dungeon object (wand)*

U+22A3

## Armor



*Armor — the blackboard A: a dungeon object (armor), first of the dungeon entity glyphs*

U+1D538

## Weapon



*Weapon — the blackboard W: a dungeon object (weapon), second of the dungeon entity glyphs*

U+1D54E

## Amulet



*Amulet — eight-petalled floret: a dungeon object (amulet). (Was U+1DF76, tofu on device — swapped 2026-10-10)*

U+2741

## Food



*Food — the blackboard F: a dungeon object (food)*

U+1D53D

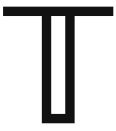
## Scroll



*Scroll — the blackboard S: a dungeon object (scroll)*

U+1D54A

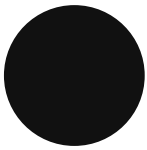
## Tool



*Tool — the blackboard T: a dungeon object (tool)*

U+1D54B

## Gem



*Gem — solid black circle: a dungeon object (gem). (Was U+2733 — swapped 2026-10-10)*

U+25CF

## Potion



*Potion — heavy exclamation: a dungeon object (potion)*

U+2757

# Dungeon: Entities

## Player



*Player — black smiling face: the dungeon's player entity. Inherits Monster plus a few extra fields Monster does not have*

U+263B

## Monster



*Monster — the blackboard M: a dungeon monster, first of the monster glyphs*

U+1D544

# Dungeon: Food

## Tin



*Tin — circled white star: a dungeon food item (tin), first of the object species*

U+272A

## RefriedBeans



*RefriedBeans — dark-encircled six: grants teleportation*

U+277B

## Spinach



*Spinach — dark-encircled seven: Popeye strength*

U+277C

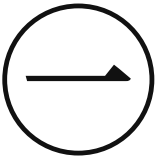
## Sushi



*Sushi — circled congratulation: the sushi/sashimi food category. (Was U+2722 — swapped 2026-10-11)*

U+3297

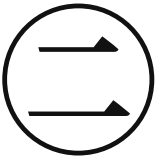
## CarpSashimi



*CarpSashimi — circled kanji: sushi item*

U+3280

## WhaleSashimi



*WhaleSashimi — circled kanji: sushi item*

U+3281

## EelSashimi



*EelSashimi — circled kanji: sushi item*

U+3282

## SalmonSashimi



*SalmonSashimi — circled kanji: sushi item*

U+3283

## TunaSashimi



*TunaSashimi — circled kanji: sushi item*

U+3284

## CrabSashimi



*CrabSashimi — circled kanji: sushi item*

U+3285

## DragonSashimi



*DragonSashimi — circled kanji: sushi item*

U+3286

## TurtleSashimi



*TurtleSashimi — circled kanji: sushi item*

U+3287

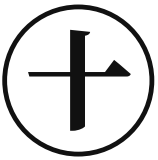
## Oyster



*Oyster — circled kanji: sushi item*

U+3288

## Uni



*Uni — circled kanji: sushi item*

U+3289

## Gyuniku



*Gyuniku — circled kanji: sushi item*

U+328A

## Asian



*Asian — squared hand: the Asian food category. (Was U+328C — swapped 2026-10-11)*

U+1F210

Ramen



*Ramen — squared cjk unified ideograph-5b57: Asian food*

U+1F211

FortuneCookie



*FortuneCookie — squared cjk unified ideograph-53cc: Asian food*

U+1F212

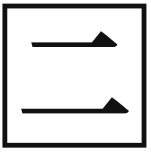
Kelp



*Kelp — squared katakana de: Asian food*

U+1F213

Dumplings



*Dumplings — squared cjk unified ideograph-4e8c: Asian food*

U+1F214

BowlOfRice



*BowlOfRice — squared cjk unified ideograph-591a: Asian food*

U+1F215

SpringRolls



*SpringRolls — squared cjk unified ideograph-89e3: Asian food*

U+1F216

Tofu



*Tofu — squared cjk unified ideograph-5929: Asian food*

U+1F217

PreservedEgg



*PreservedEgg — squared cjk unified ideograph-4ea4: Asian food*

U+1F218

## Soup



*Soup — circled twenty: the soup category*

U+2473

## Deli



*Deli — black diamond minus white X: the delicatessen food category (the 25 staples, names lost 2026-10-10 — rebuild pending)*

U+2756

## Fruit



*Fruit — domino tile back: the fruit food category*

U+1F030

## CantoneseDuck




*CantoneseDuck — squared cjk unified ideograph-6620: Asian food*

U+1F219

## Egg

Ø

*Egg — o with stroke: the egg. (Was U+2800 — swapped 2026-10-11)*

U+00F8

## Meat



*Meat — CJK beautiful: the meat category*

U+4E3D

## Creature



*Creature — o with long stroke overlay: the monster-shuffle category*

U+A74B

## Candy




*Candy — mahjong joker: the candy category*

U+1F02A

## Goodbar



*Goodbar — mahjong east wind: candy*

U+1F000

## Honee



*Honee — mahjong south wind: candy*

U+1F001

## DRation



*DRation — mahjong west wind: candy*

U+1F002

## DoveBar



*DoveBar — mahjong north wind: candy*

U+1F003

## AcidBroth

❶

*AcidBroth — dark-encircled number: soup*

U+2776

## GoldenSoup

❷

*GoldenSoup — dark-encircled number: soup*

U+2777

## StonedSoup

❸

*StonedSoup — dark-encircled number: soup*

U+2778

## CrowBar




*CrowBar — mahjong winter: candy*

U+1F029



## MoldySoup

*MoldySoup — dark-encircled number: soup*

U+2779



## Manwich

*Manwich — dark-encircled number: soup*

U+277A



## CreamOfAlgaeSoup

*CreamOfAlgaeSoup — dark-encircled number: soup*

U+277E



## ChickenNoodleSoup

*ChickenNoodleSoup — dark-encircled number: soup*

U+277F

## Apple



*Apple — domino tile: fruit, inherits Fruit*

U+1F032

## Orange



*Orange — domino tile: fruit, inherits Fruit*

U+1F033

## Pear



*Pear — domino tile: fruit, inherits Fruit*

U+1F034

## Melon



*Melon — domino tile: fruit, inherits Fruit*

U+1F035

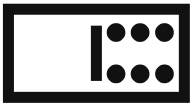
## Banana



*Banana — domino tile: fruit, inherits Fruit*

U+1F036

## Carrot



*Carrot — domino tile: fruit, inherits Fruit*

U+1F037

## Fig



*Fig — domino tile: fruit, inherits Fruit*

U+1F038

## Lemon



*Lemon — domino tile: fruit, inherits Fruit*

U+1F039

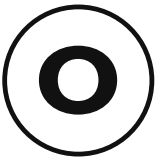
## Tomb




*Tomb — circled 10 on black square: safe-row tin, now Sundry*

U+3248

## Sundry



*Sundry — circled ideograph: the sundry/miscellaneous category*

U+3267

## Lime



*Lime — domino tile: fruit, inherits Fruit*

U+1F03A

## Grapes



*Grapes — domino tile: fruit, inherits Fruit*

U+1F03B

## Pesto




*Pesto — circled 20 on black square: safe-row tin, now Sundry*

U+3249

## KelpFrond




*KelpFrond — circled water: sundry item (kelp is seaweed)*

U+328C

## EucalyptusLeaf




*EucalyptusLeaf — circled wood: sundry item*

U+328D

## CandyBar




*CandyBar — circled metal: sundry item*

U+328E

## LembasWafer




*LembasWafer — circled earth: sundry item*

U+328F

## CramRation




*CramRation — circled sun: sundry item*

U+3290

## FoodRation




*FoodRation — circled medicine: sundry item*

U+32A9

## KRation




*KRation — circled religion: sundry item*

U+32AA

## CloveOfGarlic




*CloveOfGarlic — circled attention: sundry item*

U+329F

## SprigOfWolfsbane




*SprigOfWolfsbane — circled item: sundry item*

U+32A0

## LumpOfRoyalJelly




*LumpOfRoyalJelly — circled copy: sundry item*

U+32A2

## CRation




*CRation — circled study: sundry item*

U+32AB

# Dungeon: Species inheritance

## EarthElemental



*⊏ Elementals — Species inherits its category (2026-10-10)*

U+2295

## FireElemental



*⊏ Elementals — Species inherits its category (2026-10-10)*

U+229B

## AirElemental



*⊏ Elementals — Species inherits its category (2026-10-10)*

U+2A37

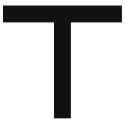
## WaterElemental



*⊏ Elementals — Species inherits its category (2026-10-10)*

U+2A38

## PurpleWorm



*⊑ Worms — Species inherits its category (2026-10-10)*

U+22A4

## BabyPurpleWorm



*⊑ Worms — Species inherits its category (2026-10-10)*

U+22A6

## LongWorm



*⊑ Worms — Species inherits its category (2026-10-10)*

U+2A53

## BabyLongWorm



*⊑ Worms — Species inherits its category (2026-10-10)*

U+2A55

## LargeKobold



*⊏ Kobolds — Species inherits its category (2026-10-10)*

U+049C

## Kobold



*⊏ Kobolds — Species inherits its category (2026-10-10)*

U+049D

## MasterMindFlayer



*⊏ Mindflayers — Species inherits its category (2026-10-10)*

U+04F4

## MindFlayer



*⊏ Mindflayers — Species inherits its category (2026-10-10)*

U+04F5

## StoneGiant

Ġ

*⊑ Giants — Species inherits its category (2026-10-10)*

U+0120

## Giant

Ģ

*⊑ Giants — Species inherits its category (2026-10-10)*

U+0122

## KoboldLord

Ќ

*⊑ Kobolds — Species inherits its category (2026-10-10)*

U+040C

## KoboldShaman

Ķ

*⊑ Kobolds — Species inherits its category (2026-10-10)*

U+049B

Ĝ

## FireGiant

⊑ *Giants — Species inherits its category (2026-10-10)*

U+011C

Ğ

## HillGiant

⊑ *Giants — Species inherits its category (2026-10-10)*

U+011E

Ǵ

## FrostGiant

⊑ *Giants — Species inherits its category (2026-10-10)*

U+0193

G ˇ

## Ettin

⊑ *Giants — Species inherits its category (2026-10-10)*

U+01E6

## VampireBat

ƀ

*⊑ Vampires — Species inherits its category (2026-10-10)*

U+0180

## StormGiant

Ǧ

*⊑ Giants — Species inherits its category (2026-10-10)*

U+01E4

## Vampire

Ỿ

*⊑ Vampires — Species inherits its category (2026-10-10)*

U+1E7F

## Titan

⅁

*⊑ Giants — Species inherits its category (2026-10-10)*

U+2141

## VampireLeader



*⊑ Vampires — Species inherits its category (2026-10-10)*

U+1E7D

## VampireLady



*⊑ Vampires — Species inherits its category (2026-10-10)*

U+2047

## VampireLord



*⊑ Vampires — Species inherits its category (2026-10-10)*

U+2047

## Nosferatu



*⊑ Vampires — Species inherits its category (2026-10-10)*

U+2047

## HumanMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## ElfMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## DwarfMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## GnomeMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## OrcMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## KoboldMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## GiantMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## EttinMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## KoboldZombie

ℤ

⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

## GnomeZombie

ℤ

⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

## OrcZombie

ℤ

⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

## DwarfZombie

ℤ

⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

## ElfZombie



*⊑ Undead — Species inherits its category (2026-10-10)*

U+2124

## HumanZombie



*⊑ Undead — Species inherits its category (2026-10-10)*

U+2124

## GiantZombie



*⊑ Undead — Species inherits its category (2026-10-10)*

U+2124

## EttinZombie



*⊑ Undead — Species inherits its category (2026-10-10)*

U+2124

## Ghoul

ℊ

*⊑ Undead — Species inherits its category (2026-10-10)*

U+210A

## WoodElf



*⊑ Elves — Species inherits its category (2026-10-10)*

U+2618

## GreenElf



*⊑ Elves — Species inherits its category (2026-10-10)*

U+2618

## GreyElf



*⊑ Elves — Species inherits its category (2026-10-10)*

U+2618

## BabyDragon

⁇

*⊑ Dragons — Species inherits its category (2026-10-10)*

U+2047

## Elf



*⊑ Elves — Species inherits its category (2026-10-10)*

U+2618

## ElfNoble



*⊑ Elves — Species inherits its category (2026-10-10)*

U+2E19

## ElvenMonarch

支

*⊑ Elves — Species inherits its category (2026-10-10)*

U+2F40

## Centipede

¢

*⊑ Spiders — Species inherits its category (2026-10-10)*

U+00A2

## CaveSpider

¤

*⊑ Spiders — Species inherits its category (2026-10-10)*

U+00A4

## Scorpion

±

*⊑ Spiders — Species inherits its category (2026-10-10)*

U+00B1

## Dragon

⁇

*⊑ Dragons — Species inherits its category (2026-10-10)*

U+2047

## GiantSpider

Ø

*⊑ Spiders — Species inherits its category (2026-10-10)*

U+00D8

## Mastodon

Ψ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+03A8

## Mumak

Ψ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+03A8

## Baluchitherium

ϒ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+03D2

## Wumpus

¿

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+00BF

## Trapper



*⊑ Beasts — Species inherits its category (2026-10-10)*

U+048A

## Minotaur

Ṁ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+1E40

## Titanothere



*⊑ Beasts — Species inherits its category (2026-10-10)*

U+213A

## Bugbear

ö

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+00F6

## Owlbear

ø

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+00F8

## Jabberwock

ƹ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+01B8

## Rothe

ȣ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+0222

## Shade

@

*⊑ Guises — Species inherits its category (2026-10-10)*

U+0040

## Leocrotta

ꝉ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+023D

## Stalker

Ǝ

*⊑ Lurkers — Species inherits its category (2026-10-10)*

U+042D

## LurkerAbove



*⊑ Lurkers — Species inherits its category (2026-10-10)*

U+214F

## Skeleton

@

⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

## Foocubus

@

⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

## MountainNymph

@

⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

## WoodNymph

@

⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

## WaterNymph

@

⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

## MerfolkRoyal

⁇

⊑ *Merfolk — Species inherits its category (2026-10-10)*

U+2047

## MerfolkNoble

⁇

⊑ *Merfolk — Species inherits its category (2026-10-10)*

U+2047

## MerfolkWitch

⁇

⊑ *Merfolk — Species inherits its category (2026-10-10)*

U+2047

## WizardOfYendor

¥

*⊑ Specials — Species inherits its category (2026-10-10)*

U+00A5

## Shopkeeper

§

*⊑ Specials — Species inherits its category (2026-10-10)*

U+00A7

## MerfolkCommon

⁇

*⊑ Merfolk — Species inherits its category (2026-10-10)*

U+2047

## GrandMaster

⁇

*⊑ Specials — Species inherits its category (2026-10-10)*

U+2047

## Tin



*⊑ Food — Tin inherits Food — first object species joins the hierarchy (2026-10-10)*

U+272A

## AcidBroth



*⊑ Soup — Soup inherits Soup (2026-10-10)*

U+2776

## GoldenSoup



*⊑ Soup — Soup inherits Soup (2026-10-10)*

U+2777

## StonedSoup



*⊑ Soup — Soup inherits Soup (2026-10-10)*

U+2778

4

## MoldySoup

*⊑ Soup — Soup inherits Soup (2026-10-10)*

U+2779

5

## Manwich

*⊑ Soup — Soup inherits Soup — the human-flesh tin (2026-10-10)*

U+277A

9

## CreamOfAlgaeSoup

*⊑ Soup — Soup inherits Soup (2026-10-10)*

U+277E

10

## ChickenNoodleSoup

*⊑ Soup — Soup inherits Soup (2026-10-10)*

U+277F

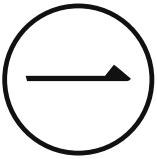
## RefriedBeans



*⊑ Soup — Soup item inherits Soup — grants teleportation (2026-10-11)*

U+277B

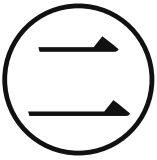
## CarpSashimi



*⊑ Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3280

## WhaleSashimi



*⊑ Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3281

## Sushi



*⊑ Food — Sushi inherits Food — the sushi food category (2026-10-10)*

U+3297

## EelSashimi



*⿸ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3282

## SalmonSashimi



*⿸ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3283

## TunaSashimi



*⿸ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3284

## CrabSashimi



*⿸ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3285

## DragonSashimi



*⿸ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3286

## TurtleSashimi



*⿸ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3287

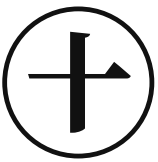
## Oyster



*⿸ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3288

## Uni



*⿸ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3289

## Gyuniku



*⊑ Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+328A

## Asian



*⊑ Food — Asian inherits Food — the Asian food category (2026-10-10)*

U+1F210

## Ramen



*⊑ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F211

## FortuneCookie



*⊑ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F212

Kelp



*⿸ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F213

Dumplings



*⿸ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F214

BowlOfRice



*⿸ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F215

SpringRolls



*⿸ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F216

## Deli



*⿸ Food — Deli inherits Food — the delicatessen food category (2026-10-10)*

U+2756

## Tofu



*⿸ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F217

## PreservedEgg



*⿸ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F218

## CantoneseDuck



*⿸ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F219

## KelpFrond



*⿸ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328C

## EucalyptusLeaf



*⿸ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328D

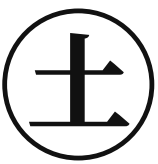
## CandyBar



*⿸ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328E

## LembasWafer



*⿸ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328F

CramRation

㊐

*㊐ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+3290

FoodRation

㊑

*㊐ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32A9

KRation

㊒

*㊐ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32AA

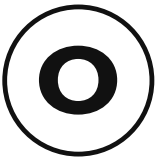
CRation

㊓

*㊐ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32AB

## Sundry



*㆓ Deli — Sundry is part of Deli (2026-10-10)*

U+3267

## CloveOfGarlic



*㆓ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+329F

## SprigOfWolfsbane



*㆓ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32A0

## LumpOfRoyalJelly



*㆓ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32A2

## Fruit



*⊑ Food — Fruit inherits Food — the fruit food category (2026-10-10)*

U+1F030

## Apple



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F032

## Orange



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F033

## Pear



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F034

## Melon



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F035

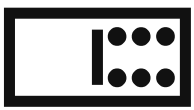
## Banana



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F036

## Carrot



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F037

## Fig



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F038

## Soup



*⊑ Tin — Soup inherits Tin — the soup category (2026-10-11)*

U+2473

## Lemon



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F039

## Lime



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F03A

## Grapes



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F03B

## Goodbar



*⊑ Candy — Goodbar inherits Candy (2026-10-11)*

`U+1F000`

## Honee



*⊑ Candy — Honee inherits Candy (2026-10-11)*

`U+1F001`

## DRation



*⊑ Candy — DRation inherits Candy (2026-10-11)*

`U+1F002`

## Candy



*⊑ Food — Candy inherits Food — the candy category (2026-10-11)*

`U+1F02A`

## Tomb



*⿺ Sundry — Tomb inherits Sundry*
*(2026-10-11)*

U+3248

## Pesto



*⿺ Sundry — Pesto inherits Sundry*
*(2026-10-11)*

U+3249

## DoveBar



*⿺ Candy — DoveBar inherits Candy*
*(2026-10-11)*

U+1F003

## CrowBar



*⿺ Candy — CrowBar inherits Candy*
*(2026-10-11)*

U+1F029

## Egg

Ø

*⊑ Creature — Egg inherits Creature (2026-10-11)*

U+00F8

## Spinach



*⊑ Soup — Spinach (NetHack tin filling, Popeye strength) inherits Soup (2026-10-11)*

U+277C

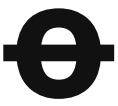
## Meat



*⊑ Creature — Meat inherits Creature — the meat category (2026-10-11)*

U+4E3D

## Creature



*⊑ Tin — Creature inherits Tin — the creature-meat tin category; the category that shuffles monsters as consumables (tins of monster meat) (2026-10-11)*

U+A74B



## IntegerBounded

⊑ *Integer — IntegerBounded inherits Integer (new 2026-10-15)*

U+012A



## IntegerBounded.max

⊑ *Integer — IntegerBounded.max is an Integer (new 2026-10-15)*

U+2047



## IntegerBounded.min

⊑ *Integer — IntegerBounded.min is an Integer (new 2026-10-15)*

U+2047



## IntegerBounded.epsilon

⊑ *Integer — IntegerBounded.epsilon is an Integer (new 2026-10-15)*

U+2047

## IntegerBounded.nabla

⁇

*IntegerBounded.nabla enforces bounded gets and sets (new 2026-10-15)*

U+2047

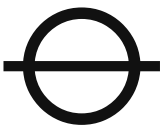
## IntegerBounded.nabla

⁇

*⊑ Interpreter — IntegerBounded.nabla is an Interpreter (new 2026-10-15)*

U+2047

## Angle



*⊑ IntegerBounded — Angle inherits IntegerBounded (new 2026-10-15)*

U+29B5

## VirtualHorizontal



*⊑ Angle — VirtualHorizontal inherits Angle (new 2026-10-15)*

U+29D3

## Egg

Ø

*Egg stays empty till use — populated by the Creature monster-shuffle at runtime (2026-10-11)*

U+00F8

## VirtualDistance



*⊑ IntegerBounded — VirtualDistance inherits IntegerBounded (new 2026-10-15)*

U+27F0

## VirtualVertical



*⊑ Angle — VirtualVertical inherits Angle (new 2026-10-15)*

U+29D7

## Meat



*Meat stays empty till use — populated by the Creature monster-shuffle at runtime (2026-10-11)*

U+4E3D

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